Gautam Kanna Kannapiran

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience delivering production-grade machine learning, Generative AI, and agentic AI solutions across fintech and healthcare. Expertise in fraud detection, transaction risk intelligence, explainable ML, clinical NLP, and RAG using Python, PyTorch, XGBoost, and transformer models. Skilled in building secure, scalable AI applications with hybrid retrieval, LLM evaluation, prompt-injection defense, and PII protection. Experienced in deploying ML models and microservices across AWS and GCP using Docker, MLflow, CI/CD, and model monitoring to improve reliability, scalability, and operational efficiency.

TECHNICAL SKILLS

Programming Languages: Python, SQL, TypeScript, JavaScript

Machine Learning: PyTorch, Scikit-learn, XGBoost, Classification, Regression, Anomaly Detection, Risk Scoring, Feature Engineering, Ensemble Learning, Hyperparameter Optimization, Model Calibration, Explainable AI (SHAP)

Generative AI & LLMs: GPT-3, Transformer Models, Prompt Engineering, Agentic AI, LLM Guardrails, Prompt-Injection Defense, PII Redaction, LLM Evaluation, LLM-as-a-Judge, Text-to-SQL

NLP & Information Retrieval: BERT, spaCy, Named Entity Recognition (NER), Clinical NLP, Text Summarization, Retrieval-Augmented Generation (RAG), Hybrid Search, BM25, Semantic Search, Cross-Encoder Reranking

LLM Frameworks & Vector Search: LangChain, FAISS, pgvector, Vector Embeddings, Similarity Search, Semantic Caching

MLOps & Data Engineering: MLflow, Apache Airflow, Docker, CI/CD, Model Versioning, Model Monitoring, Data Validation, Drift Detection, GitHub Actions

Cloud & Deployment: AWS SageMaker, Amazon S3, Amazon CloudWatch, Google Cloud Platform (GCP), Google Compute Engine (GCE), Google Cloud Storage (GCS), Artifact Registry, FastAPI, Python Microservices, Azure

Databases & Data Security: PostgreSQL, MySQL, MongoDB, SQL Server, Role-Based Access Control, OAuth 2.0, PII Protection

Business Intelligence: Power BI, Power Query, DAX, Tableau

WORK EXPERIENCE

AI/ML Engineer | Stripe, Canada **Jan 2024 - Present**

- Developed an agentic GenAI fraud investigation and risk intelligence platform automating alert triage, explainable case summaries, and natural-language analysis of transaction, chargeback, and AML/KYC data, reducing investigation time by **40%**.
- Trained an **XGBoost** risk-scoring model on engineered velocity, device, and merchant features, serving real-time scores with **SHAP** explanations to prioritize high-risk alerts and reduce false positives.
- Implemented **LLM** guardrails for intent classification, PII redaction, and prompt-injection defense with per-tenant model routing, optimizing chain-of-thought/few-shot prompts via LLM-as-a-judge evals, reducing fraud escalations by **35%**.
- Built a schema-aware text-to-SQL engine using **PostgreSQL** metadata introspection, enabling fraud analysts to query transaction and dispute data in natural language with tenant-isolated, RBAC-enforced execution.
- Designed a hybrid RAG pipeline with **LangChain** and **TypeScript pgvector**, combining BM25, dense retrieval, and cross-encoder reranking, improving retrieval precision by **28%** and cutting token costs by **32%** via semantic caching.
- Automated **Apache Airflow ETL pipelines** ingesting **2M+** daily transactions and regulatory PDFs from **Google Cloud Storage (GCS),** orchestrating boundary-aware chunking, embedding refreshes, **pgvector** re-indexing, and data-quality checks.
- Deployed containerized **Python microservices** on **GCE** using **Artifact Registry** and **GitHub Actions CI/CD**, integrating automated testing and LLMOps observability to improve release velocity and reliability across environments.
- Contributed to a production LLM orchestration platform enabling self-service fraud and risk intelligence for **45+** risk, compliance, and operations teams, reducing time-to-market for new automation use cases by **60%**.
- Collaborated with fraud analysts, data scientists, and compliance stakeholders in **Agile/Scrum**, translating investigation requirements into production GenAI features through sprint planning, code reviews, and **Jira** backlog grooming.

ML Engineer | McKesson Corporation, Canada **Jun 2021 – Dec 2023**

- Spearheaded analytics across **150K+ de-identified records**, standardizing enrollment, reimbursement, prescription fulfillment, and treatment data through quality checks, missing-value handling, and longitudinal feature engineering.
- Formulated temporal risk-classification models using **PyTorch** and **Scikit-learn** to predict delayed therapy initiation and potential discontinuation, achieving **84% precision** through threshold calibration and **SHAP**-based explainability.
- Advanced clinical NLP using **BERT** and **spaCy** to extract diagnoses, medication regimens, authorization requirements, and treatment milestones from **3 document types**, applying terminology normalization and annotated-data validation.
- Engineered a **FAISS** semantic retrieval pipeline indexing **100K+ clinical and policy passages**, combining metadata filtering and relevance reranking to retrieve treatment guidance and payer authorization criteria with **sub-500ms** latency.
- Delivered a human-reviewed clinical summarization workflow using **GPT-3** and **LangChain** to generate evidence-linked case summaries from authorization and patient-support records, preserving source traceability and flagging unsupported outputs.
- Refined model generalization through **Optuna Bayesian optimization**, temporal cross-validation, and **Monte Carlo** stress testing, evaluating precision, macro **F1-score**, class imbalance, missing treatment events, and population shifts.
- Operationalized inference on **AWS SageMaker** with **MLflow**, **GitHub Actions CI/CD**, and **Amazon CloudWatch**, establishing model-version tracking, deployment validation, latency monitoring, and drift alerts for controlled production releases.
- Instituted clinical model oversight with pharmacists and care teams, applying **PHIPA/PIPEDA** safeguards and delivering **Power BI** dashboards using **Power Query** and **DAX measures** to track model performance and patient-support outcomes.

PROJECTS

Cross-Sectional Cryptocurrency Momentum and Reversal Research (Python, Pandas, NumPy):

- Built a Python market-neutral crypto strategy, ranking 10 major coins for long-short positioning.
- Pulled 4-hour market data from the Binance API with python-binance and cached it to CSV for reproducibility.
- Modeled transaction costs (20 bps taker, 7 bps maker) and exponentially smoothed portfolio weights to reduce turnover. Achieved a 0.51 Sharpe ratio, −13.9% max drawdown, & −0.06 BTC correlation on 2024–2025 out-of-sample data after costs.

Quant portfolio website:

- Built a responsive single-page portfolio site in plain HTML, CSS, and JavaScript with no frameworks or build step.
- Implemented a light/dark theme toggle that persists the visitor's choice, plus mobile navigation and scroll-reveal animations.
- Designed a custom CSS design system with theme tokens for both light and dark modes.
- Published the site on GitHub Pages, with automatic redeploys on each push.

Chess Application (Java):

- Built a two-player chess game in Java using Swing, with no external libraries.
- Implemented all standard rules, including castling, en passant, pawn promotion, check, checkmate, and stalemate.
- Designed a modular architecture with separate Piece, Board, ChessGame, and Main classes.
- Validated legal chess moves through simulation, preventing king exposure and handling pins without special-case logic.

Expense Tracker (Java):

- Developed a command-line expense tracker in Java with input validation for amounts and dates.
- Built persistent storage that saves every change to a CSV file, so data survives program exits.
- Implemented spending analytics, including totals, category breakdowns, and a date-sorted running total.
- Applied separation of concerns across Expense, ExpenseTracker, and Main classes, with non-reused IDs for deletions.

CERTIFICATIONS

- **Wall Street Quants: Quant Bootcamp**

EDUCATION

Bachelor of Science, Computer Science: York University, Toronto, Ontario, Canada.